

Kevin Jiang

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Education

University of Waterloo

Bachelor of Software Engineering (Honours)

Sep 2024 – Apr 2029

Experience

Software Engineer - Geospatial AI Intern, Epoch Blue | London, UK

Sept 2026 - Dec 2026 | [About](#)

- Productionized a **global 60°S–80°N dataset** for groundwater and flood models using scalable geospatial ETL pipelines
- Built **concurrency-safe** PostgreSQL checkout and transaction-recovery systems, ensuring **atomic billing workflows**
- Validated **seven** mirrored **global datasets**, documenting COG conversion pipelines and Icechunk access infrastructure

Full-Stack Developer, Royal Canadian Air Force (RCAF) | Ottawa, ON

Jan 2026 - Apr 2026 | [About](#)

- Built and **deployed two web applications** for RCAF operations, streamlining workflows and improving access to data
- Implemented RBAC in AWS Cognito whilst incorporating **domain-driven design**, enhancing security with modularity
- Designed and tested **50+ RESTful API** endpoints and **15+ schemas** in PostgreSQL powering core app features
- Built **10+ webpages** and components using Vue.js, collaborating with UI/UX designers for user-centric design
- Reviewed **25+ pull requests**, improving code quality, correctness, and data consistency

Data Engineering Research Assistant, University of Waterloo | Waterloo, ON

May 2025–Present

- Aggregated **10.7 million** geospatial data points from **22 sources** into PostgreSQL databases using ArcGIS and Python
- Built and documented **5 modules** in R/Python for data analysis use with the **Climate Risk Research Group**
- Developed **6 web dashboards** using React, R Shiny, and RMarkdown to visualize assorted flood data
- Prototyped approaches integrating novel data and techniques into **flood loss models**, evaluating impact on accuracy

Software Engineer Intern, ReliablyME | Toronto, ON

Jan 2025 - Apr 2025 | [Github](#)

- Black-box tested **2400+ lines** of RESTful API code used in AI-driven commitment tracker startup, improving reliability
- **Implemented features** allowing users to backdate or update their dates of commitment completion

Projects

1st Place, Factualial @ SOA Research Challenge | 68 teams | R, Python, NumPy

Mar 2025 | [Paper](#) | [GitHub](#)

- Awarded **\$10000** for designing a national flood insurance program with policy experts using **layered risk modeling**
- Built **machine learning pipelines** using Google's YDF for EDA, parameter tuning, and model validation in R/Python
- Featured on [SOA podcast](#), [The Actuary](#), [ImpACT Conference](#), and the [UWaterloo Math Department](#)

2nd Place, UTRAHacks 2025 | 249 participants | C++, Arduino, Robotics

Jan 2025 | [GitHub](#)

- Built an Arduino-based robotic control systems for a robotics navigation and path-finding competition
- Programmed C++ **state machine algorithms** to complete path-finding challenges

3rd Place, Debug Debacle @ McHacks 2025 | 471 participants | Python, React.js, SocketIO

Jan 2025 | [GitHub](#) | [Link](#)

- Published **full-stack multiplayer game** with OAuth and Vercel where players race to fix AI-generated bugged code
- Wrote **code validation scaffolds** to test if player submissions correctly solved GumloopAI AI-generated bugs

Firmware Member, Waterloop | C++, STM32, CAN bus protocol

Sep 2024 - Present | [Github](#)

- Integrated new Raspberry Pi as central control unit of the design team's machine via **CAN protocol**
- Developed and documented **STM32 scripts** using C++ to verify proper operation of the CAN network

Skills and Qualifications

- **Languages:** Python, R, SQL, TypeScript/JavaScript, C/C++, HTML/CSS, Bash
- **Frameworks & Libraries:** React, Vue, FastAPI, Flask, NumPy, Pandas, GeoPandas, Xarray, Dask, PyTorch, TensorFlow
- **Data & Geospatial:** PostgreSQL/PostGIS, BigQuery, Google Earth Engine, Zarr, Cloud-Optimized GeoTIFFs (COGs), Icechunk, ArcGIS/QGIS, Leaflet, MapboxGL
- **Cloud & Infrastructure:** Docker, Terraform, AWS, GCP, Git, Linux, Node.js